

Estela Pérez Luque, PhD

Biomechanics Product Specialist

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PROFILE

Biomechanics specialist with extensive hands-on experience designing and executing end-to-end motion capture workflows across automotive applications and others. I have worked with optical and wearable mocap systems, EMG systems, pressure platforms, and simulation tools to validate measurement pipelines and improve data quality. I bring structured testing, cross-functional collaboration, and a strong problem-solving mindset to product teams developing advanced motion capture solutions. I am motivated to contribute to Qualisys' biomechanics product development by bridging user needs with robust technical validation.

TECHNICAL SKILLS

Motion Capture & Biomechanics: motion capture systems (Xsens, MOVE4D, optical), EMG, force plates, multi-modal system synchronization, biomechanical data collection, time-series kinematic analysis, end-to-end biomechanics workflows.

Data Processing & Validation: Experimental design, protocol development, data quality control, motion data processing, model validation, multi-system data integration, real-world dataset analysis.

Programming & Modelling: MATLAB, Python; Predictive modelling; Inverse kinematics; Simulation-based analysis; Optimisation methods.

Tools: Digital Human Modelling tools: IPS IMMA, Siemens Jack, Anybody; CAD: Autodesk Inventor, SolidWorks.

Languages: English (C1), Spanish (Native), Swedish (B1).

PROFESSIONAL EXPERIENCE

PhD Researcher, University of Skövde (Sweden)

Sep 2020 – Oct 2025

- **Motion Capture Workflows and Third-Party Integration:** I contributed to the data collection processes and analysis across multiple applications (dancers, athletics, factory workers) by combining different motion capture systems with wearable sensors, including multi-system synchronization, resulting in high-quality datasets for biomechanical analysis and modelling.
- **Structured Problem-Solving, Collaboration, and Communication:** I improved posture and motion prediction methods in simulation tools by collaborating with industry partners (Volvo Cars, Volvo Trucks, Scania) and research institutes (Texas Tech University, University of Antwerp), translating user needs into testable study designs and executing experimental protocols end-to-end (design, data collection, processing, analysis, reporting). I communicated results through publications and stakeholder-focused deliverables, enabling the adoption of more accurate and robust biomechanics workflows.

Visiting Researcher - University of Antwerp (Belgium)

Sep 2024 – Mar 2025

- **Cross-Functional Coordination, Study Governance, and Data Pipelines:** I led a six-month international research stay and managed the technical preparation across institutions. I took full ownership of ethics applications and experimental protocol design for motion-capture, and I built structured, reproducible data pipelines for advanced 4D body-scanning and motion-capture data. I delivered reliable workflows and documentation that enabled multi-site collaborators to run consistent, data-driven validation and produce high-quality outputs.

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Co-founder and R&D Lead - Swe-Ex AB (Startup) (Sweden)

Sep 2021 – Mar 2023

- **Product Development and Prototype Validation:** As R&D Lead, I co-founded a wearable exoskeleton startup and directed concept development and prototyping with a focus on user safety and ergonomics. I translated user requirements into testable design targets under real-world constraints, and I drove the iteration cycle from CAD to physical build using Autodesk Inventor and SolidWorks, producing three functional prototype iterations with 3D-printed, textile, and steel components. I validated core functionality through prototype testing and stakeholder demonstrations, showing end-to-end ownership from concept through physical verification.

Research Assistant - University of Skövde (Sweden)

Jun 2019 – Sep 2020

- **Product demonstration and Communication:** I supported applied ergonomics projects with 6+ industry partners by conducting analysis, producing decision-ready reports and educational documentation (including video content), and presenting findings to cross-functional teams. I helped partners apply ergonomic insights in product development discussions and translate findings into practical deliverables.

EDUCATION

PhD in Informatics, University of Skövde (Sweden)

Sep 2020 – Oct 2025

[PhD Thesis](#): Human Posture and Motion Prediction for Automotive Ergonomics Design.

Focus: motion capture data validation of posture and motion prediction, workflow development, and simulation accuracy.

Master of Science in Intelligent Automation, University of Skövde (Sweden)

Aug 2018 – Jun 2019

Master thesis: Experimental evaluation of three upper-body exoskeletons using Xsens motion capture (study design, data collection, and analysis).

Bachelor of Science in Mechanical Engineering, University of Skövde (Sweden)

Aug 2017 – Jun 2018

Degree in Mechanical Engineering, University of Málaga (Spain)

Aug 2012 – Jun 2018

RESEARCH PUBLICATIONS

Full list available at: [Google Scholar – Estela Pérez Luque](#)